AUTISM SPECTRUM DISORDER CLASSIFICATION USING GRAPH NEURAL NETWORKS FROM GUT MICROBIOTA PROFILES

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The gut-brain axis can define the emotions, decisions and actions of a human being. The microbiome is an influential part in our lives. Conventional machine learning methods represent metagenomic profiles as a matrix, leaving behind the relationships and valuable characterization data. Graph Neural Networks (GNN) offer an alternative by representing patients and taxa as interconnected entities to learn directly from this structure. This study uses GNNs for the classification of Autism Spectrum Disorder (ASD) and neurotypical controls using gut microbiome profiles, identifying the different taxa with more relevance in the neurodevelopmental disorder. We used public datasets that have 123 prepubescent patients from China selected by age gap, origin, and drug exposure. Samples were previously analyzed via Illumina sequencing of 16S rRNA V4 region amplicons. Data were processed using QIIME 2 to obtain genus-level taxonomic abundance profiles. A bipartite patient-taxon graph was constructed, where nodes represent patients and taxa, and edges represent taxonomic abundances. Graph Attention Network (GAT) was implemented over this structure to classify individuals. The dataset was split into training and test sets 80:20 using node masking. The model achieved an overall accuracy of 0.84 on the test set. For neurotypical controls, precision was 0.88, recall 0.70, and F1-score 0.78, for ASD, values were 0.82, 0.93, and 0.88, respectively. Analysis via GNNExplainer allowed ranking taxa contributions and identified a subset of nodes with high influence on classification, revealing that the most influential for ASD are: Micrococcaceae, Lachnospiraceae_UCG-003 and Bacillus. Representing microbiota profiles via bipartite graphs combined with attention neural networks is a promising strategy for classifying neurodevelopmental phenotypes. Incorporating explainable AI methods identifies the taxonomic features supporting predictions, enhancing the biological interpretability of the model.